

DIRECTIONS for questions 1 to 5: Answer the questions on the basis of the information given below.

Each of the three students - Abhijit, Dheeraj and Mohit - wrote exams on three subjects - Maths, Physics and Chemistry. After the results are out, it is found that:

- i. two students passed in Maths, one in physics and one in Chemistry.
- ii. Dheeraj passed in Maths but failed in Chemistry.
- iii. Mohit failed in Maths and Abhijit failed in Physics.
- iv. every student passed in at least one subject.

Q1. DIRECTIONS for questions 1 to 5: Select the correct alternative from the given choices.

If both the students who failed in Physics also failed in Chemistry, then which of the following statements is **definitely true**?

- ☐ a) **Dheeraj passed in Physics.**
- ☐ b) Mohit passed in Maths.
- ☐ c) **Mohit failed in Chemistry.**
- ☒ d) **Abhijit failed in Chemistry.** ✓ Your answer is correct

Q2. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

If Mohit passed in Physics, then which of the following statements ***could be true***?

- ☒ a) **Each of Dheeraj and Mohit passed in exactly one subject.** ✓ **Your answer is correct**
- ☐ b) Each of Mohit and Abhijit passed in exactly one subject.
- ☐ c) **Dheeraj passed in exactly two subjects.**
- ☐ d) **Dheeraj failed in Maths.**

Q3. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
Which of the following statements *could be true*?

- ☐ a) **Both Dheeraj and Mohit failed in Physics.**
- ☐ b) Both Mohit and Abhijit passed in Maths.
- ☐ c) **Both Mohit and Abhijit failed in Chemistry.**
- ☒ d) **Dheeraj passed in exactly one subject and Mohit passed in exactly two subjects.**

Q4. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices. If Mohit passed in exactly two subjects, which of the following statements is ***definitely true***?

- ☐ a) **Dheeraj passed in Physics.**
- ☐ b) Mohit failed in Chemistry.
- ☒ c) **Abhijit failed in Chemistry.** ✓ **Your answer is correct**
- ☐ d) **Mohit failed in Physics.**

Q5. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

If one of the three persons passed in the same subjects as one of the other two persons, then which of the following statements is ***definitely true***?

- ☐ a) Mohit passed in exactly one subject.
- ☐ b) Dheeraj passed in Physics.
- ☐ c) Mohit failed in Physics.
- ☒ d) Abhijit passed in exactly one subject. ✓ Your answer is correct

DIRECTIONS for questions 6 to 10: Answer these questions on the basis of the information given below.

On a particular day, exactly twelve customers visited a beauty salon and each customer availed of exactly one service at the salon. Of the twelve, four availed of a haircut, four availed of a massage and four availed of a facial. The twelve customers entered the salon at the same time and each customer left immediately after he had availed of his respective service.

In the salon, each of the three services was provided by a different beautician – John gives haircuts, Jack gives massages and Gary gives facials. Each beautician can service only one customer at any time. The times taken by John, Jack and Gary to service one customer is between 11 minutes to 15 minutes (both inclusive), between 16 minutes to 20 minutes (both inclusive) and between 21 minutes to 25 minutes (both inclusive) respectively.

A beautician always starts servicing a customer waiting to avail of his services without delay, unless he is already servicing another customer. The time spent by any customer in the salon is the sum of the time spent by him waiting to be serviced and the time spent by the corresponding beautician in servicing him.

Given below is the total time spent by each of the twelve customers in the salon (in the alphabetical order of their names).

Person	Time spent (minutes)
Akhil	13
Anant	52
Das	36
Gautam	67

Person	Time spent (minutes)
Govind	22
Hardip	17
Jayant	43
Narendra	24

Person	Time spent (minutes)
Neeraj	50
Nitin	69
Ram	33
Shiv	90

Q6. DIRECTIONS for question 6: Select the correct alternative from the given choices.

Who among the following was serviced by Jack?

- ☐ a) Das
- ☒ b) Anant ✔ Your answer is correct
- ☐ c) Neeraj
- ☐ d) Gautam

Q7. DIRECTIONS *for question 7: Type in your answer in the input box provided below the question.*

What is the maximum time (in minutes) that John took to service any customer?

Q8. DIRECTIONS *for questions 8 and 9:* Select the correct alternative from the given choices.

Who among the following was not serviced by John?

☐ a) **Akhil**

☐ b) **Neeraj**

☐ c) **Das**

☒ d) **Ram** ✓ **Your answer is correct**

Q9. DIRECTIONS *for questions 8 and 9:* Select the correct alternative from the given choices.

Who among the following spent the maximum time waiting to get serviced in the salon?

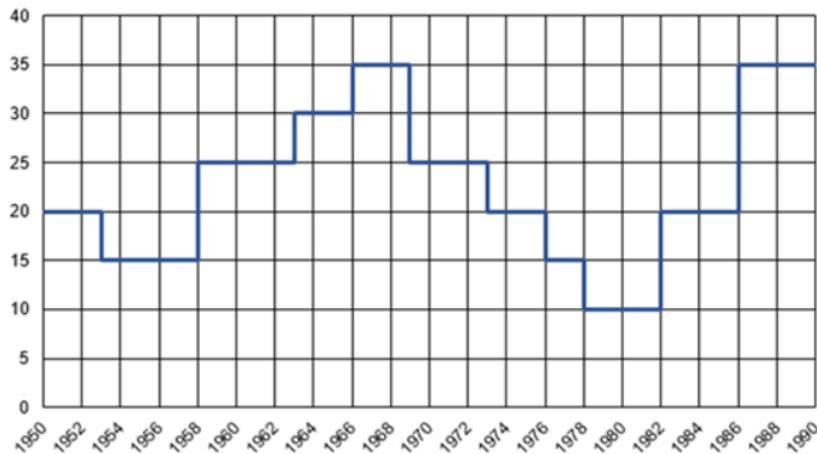
- ☐ a) **Neeraj**
- ☐ b) **Anant**
- ☐ c) **Jayant**
- ☐ d) **Das**

Q10. DIRECTIONS *for question 10: Type in your answer in the input box provided below the question.*

What is the average time (in minutes) that Gary took to service a customer?

DIRECTIONS for questions 11 to 15: Answer these questions on the basis of the information given below.

The population of Geometric Tortoise, a rare species of tortoise, in a zoological park was monitored from 1950 to 1990 (both inclusive). Their population was determined at the beginning of each year. The following chart presents this data:



Assume that the population of the tortoises remained constant during the year and changed only at the beginning of each year. For example, the population of the tortoises throughout 1958 was 25.

Q11. DIRECTIONS for question 11: Type in your answer in the input box provided below the question.

From 1951 to 1990, in how many years did the population of the tortoises decrease by more than 20% as compared to the previous year?

Q12. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.

In which year was the percentage increase in the population of the tortoises as compared to the previous year the highest?

☐ a) 1958

☐ b) 1982

☐ c) 1986

☐ d) 1963

Q13. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.

What is the average annual population of the tortoises across the given period?

☐ a) **21.78**

☐ b) **22.93**

☐ c) **23.5**

☐ d) **21.57**

Q14. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.

For the given period, if x is the number of years in which the population of the tortoises was greater than 20, what is the value of x as a percentage of the total number of years given?

- ☐ a) **48.8%**
- ☐ b) **50%**
- ☐ c) **51.3%**
- ☐ d) **46.7%**

Q15. DIRECTIONS *for question 15:* Type in your answer in the input box provided below the question.

What is the maximum number of tortoises that would have survived for the entire duration given?

DIRECTIONS *for questions 16 to 20:* Answer the questions on the basis of the information given below.

Eight students – A, B, C, D, E, F, G and H – wrote an exam conducted by an assistant professor, Asim. Asim ranked the students in the decreasing order of their marks in the exam, such that two or more students who get the same mark will get the same rank (say N) and the student who has the next highest mark will get the next rank (i.e., $N + 1$) and so on.

Later, Asim submitted the same to his superior, professor Ganesh. The professor was not satisfied and asked Asim to give revised ranks, from 1 to 8, to the eight students in the decreasing order of their marks. If two or more students get the same mark and compete for the same rank (say N), then the student with the least number of attempts among them should be ranked the best (i.e., N), the one with the next least number of attempts, the next best (i.e., $N + 1$) and so on. When Asim ranked the students accordingly, it was observed that

- i. of the students whose ranks were changed, the change in the rank of B was less than that of any other student and the change in the rank of G was more than that of any other student.
- ii. only D's rank changed by 3.
- iii. C's revised rank was not better than that of A, which, in turn, was 6.
- iv. initially, D and E had the same rank.
- v. initially, there was no rank which was given to more than three students.
- vi. no two students had the same number of attempts.

Note: Between any two ranks, the numerically lower rank is considered to be the better rank.

Q16. DIRECTIONS *for question 16:* Type in your answer in the input box provided below the question.

What is the revised rank of D?

Q17. DIRECTIONS *for question 17:* Select the correct alternative from the given choices.

Who are the students who initially got the 1st rank?

- ☐ a) **H and F**
- ☐ b) **B, D and E**
- ☐ c) **B, H and F**
- ☐ d) **D, E and H**

Q18. DIRECTIONS *for question 18:* Type in your answer in the input box provided below the question.

What is the change in the rank of C?

Q19. DIRECTIONS *for questions 19 and 20:* Select the correct alternative from the given choices.

Who got the revised rank of 1?

- ☐ a) **B**
- ☐ b) **F**
- ☐ c) **E**
- ☐ d) **Cannot be determined**

Q20. DIRECTIONS *for questions 19 and 20:* Select the correct alternative from the given choices.

Who among the following definitely cannot be the student with the least number of attempts?

☐ a) **H**

☐ b) **F**

☐ c) **D**

☐ d) **E**